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Autonomous Arduino Car with Object Detection and Avoidance

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We would also like to convey our gratitude to Islington College for providing the resource for the completion of this report and also want to thanks our Shishir sir for reviewing our group project while we were facing problem in running our projcet.

Abstract

The whole research was directed toward the design and construction of an Arduino-based autonomous car-type vehicle that can determine the nearest object in its vicinity and act on it. The vehicle is fitted with ultrasonic sensor that looks around and measures the distances from the objects in its vicinity. Based on the sensor data, the Arduino microcontroller identifies which object is the nearest to the sensor and accordingly moves the car away from that object using some decision algorithms. Four TT motors are present on the car controlled via a motor driver (L293D), which are assigned to operate forward, backward, and turn functions, respectively.

Experimental results indicate that the system detects and avoids running objects very well, thus opening arenas for some interesting educational and prototyping activities in robotics and small-scale autonomous vehicles. Currently in its infancy, future enhancements may consider employing other sensors like infrared, LIDAR, etc., for enhanced object recognition and path planning. The abstract contains contributions from many Arduino-based obstacle avoiding car projects, dominantly those that use ultrasonic sensors to measure distance and servo motors to scan.

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1. Introduction

The term IoT, or Internet of Things, refers to the whole network of linked devices as well as the technology that makes it easier for devices to communicate with one another and with the cloud (AWS, 2025). The IoT integrates “Things” with the internet using several technologies and they are Sensors and Actuators, Connectivity technologies, Cloud computing. Big data analytics, Security and privacy technologies ,etc (IBM, 2023).

Based on these principles, The Autonomous Car with Object Detection and Avoidance is a compact and practical representation of IoT in action. Using an Arduino microcontroller as its central component, this project incorporates motor drivers, control circuits that allow for autonomous decision-making and ultrasonic sensors for obstacle detection. Without any outside assistance, the vehicle is building a self-navigating car which can recognize obstacles and avoid them. This device demonstrates the concepts of autonomous mobility in a scalable and economical way by combining sensors, microcontrollers and motor control systems.

Moreover, this project demonstrates how the IoT has the potential to revolutionize smart mobility, logistics and transportation. It provides insight into how sensor fusion, control algorithms and data processing work in self-driving technologies by modelling fundamental autonomous navigation. These initiatives open the door for increasingly sophisticated, intelligent systems as IoT develops.

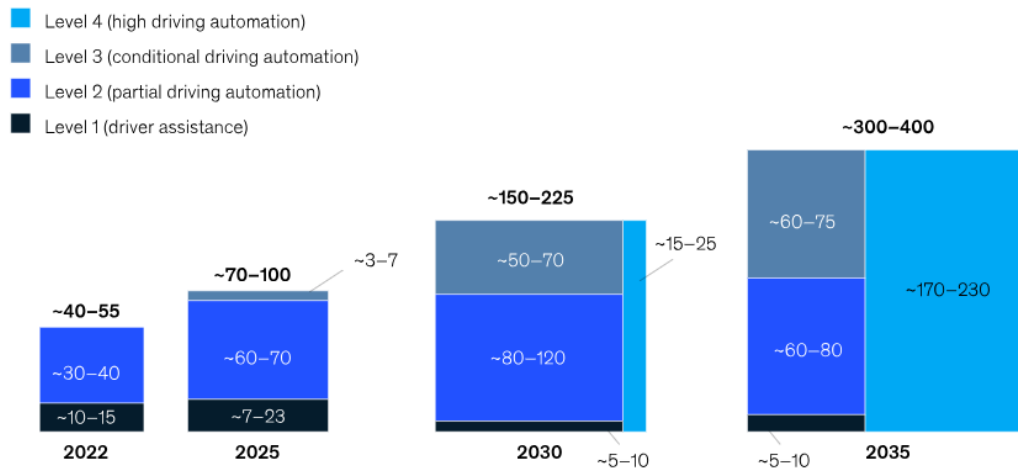
1.1. Current Scenario of Autonomous Vehicle Technology

1.1.1. International Perspective

Globally, autonomous vehicles technology has made remarkable progress with prominent companies like Waymo running robotaxi services in the US and Baidu Apollo in China which deploy hundreds of autonomous buses (Mccracken, 2025).According to McKinsey, autonomous driving could create \$300 billion to \$400 billion in revenue by 2035 (Johannes Deichmann, 2023). Although Level 4 autonomous systems are being tested in

developed countries, the cost of the technology prevents their broad use in developing countries.

Advanced driver-assistance systems (ADAS) and autonomous-driving (AD) revenues, \$ billion



Source: McKinsey Center for Future Mobility

Figure 1: ADAS/AD Revenue Forecast by Automation Level(2022-2035)

1.1.2. National Perspective

In Nepal, autonomous vehicle development has been remained in its infancy. The country is facing some challenges which includes unpredictable road conditions, lack of testing facilities and limited IoT infrastructure. However, in alignment with its second NDC goals , Nepal has aims to achieve 90% of private passenger vehicles and 60% of all four-wheeler's public passenger vehicle sales by 2030 (Paudel, 2025). This is the mean time for the Government of Nepal to think through what AV means for the future of our country.

1.2. Problem statement and project as a solution

In Nepal, the key challenges faced during the adoption of autonomous navigation system includes high costs, sensor accuracy limitations and adaptability in dynamic situations. Present-day obstacle-avoiding robots are problematic for small-scale applications due to lack of real-time responsiveness or reply on pricey sensors , high-end ultrasonic modules etc.

Our project can address these issues by replacing Arduino Uno in place of expensive processors and implementing an efficient algorithm. By designing low-power operations. This provides a foundation for future projects and proves that low-budget automation is also feasible.

1.3. Aim and Objective

- **Aim**

The main aim of this project is to design an autonomous Arduino-based car which can detect obstacles and make intelligent navigation decisions.

- **Objectives**

The main objectives are to :

- Build the car chassis with motor driver shield, wheels, tt motor, battery, Arduino and ultrasonic sensors which detect obstacles.
- Explain the software logic and algorithms implemented on the Arduino for obstacle detection and autonomous navigation.
- Implement intelligent navigation for autonomous obstacle avoidance.
- Evaluate response time and reliability in different scenarios

2. Background

2.1. System Overview

The IoT-Enhanced Autonomous Arduino Car is a self-navigating vehicle that can recognize and navigate clear of obstacles without guidance from a human. To enable autonomous activity in a scalable and efficient way, the system integrates crucial Internet of Things components such as motor drivers, motors, ultrasonic sensors, and an Arduino microcontroller. The vehicle continually tracks its immediate surroundings for impediments using ultrasonic sensors, then determines the best course of action to prevent collisions. This feature guarantees safe and efficient operation in a variety of contexts.

Regarding performance, the system uses integrated software operating on the Arduino board to process immediate information from ultrasonic sensors to estimate distances to objects ahead. The microprocessor utilizes obstacle avoidance algorithms and enables the motor drivers to automatically control the wheels' speed and direction based on the sensor facts. The chassis that holds the hardware parts are installed is made to be both mobile and balanced. The car can drive itself over a predetermined region while responding dynamically to obstacles caused by the mix of sensory input, statistical processing, and motor control.

2.2. Design Diagram

2.2.1. System Architecture

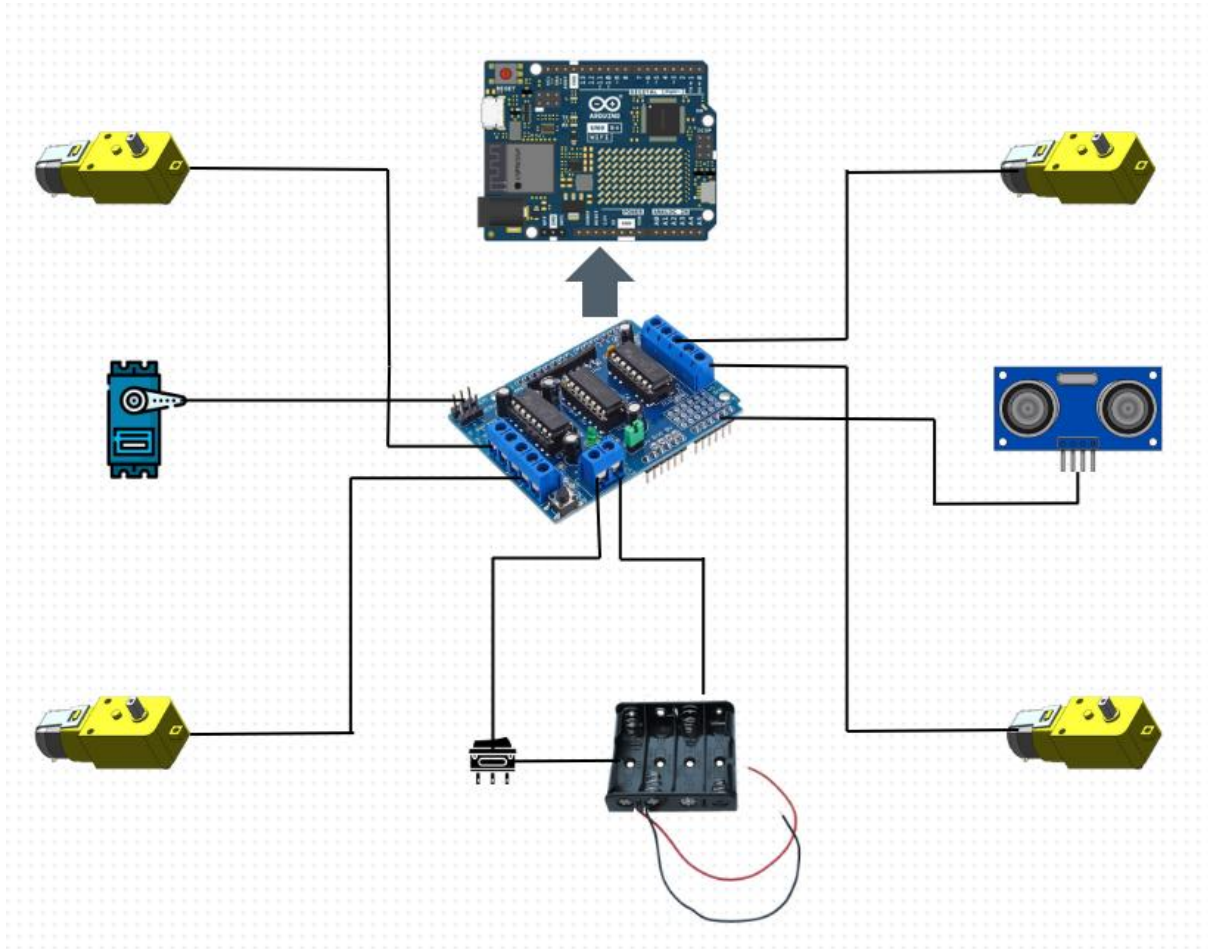


Figure 2: System architecture

2.2.2. Flowchart

This flowchart shows the process of obstacle avoidance of cars.

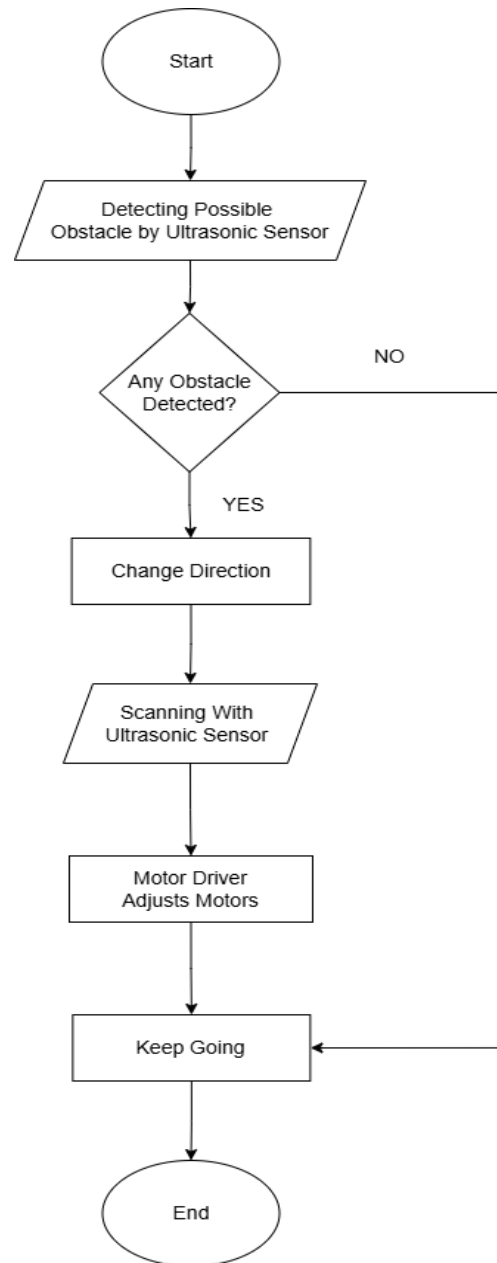


Figure 3: Flowchart of the process of obstacle avoiding car

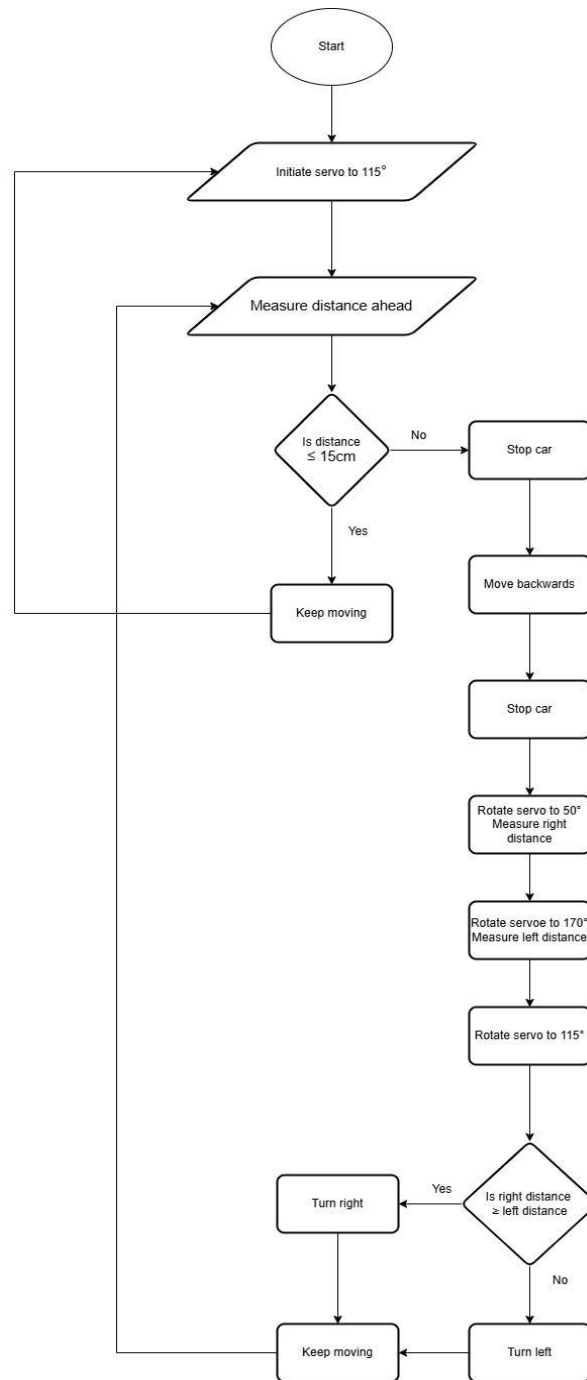


Figure 4:Flowchart based on code

2.2.3. Circuit Diagram

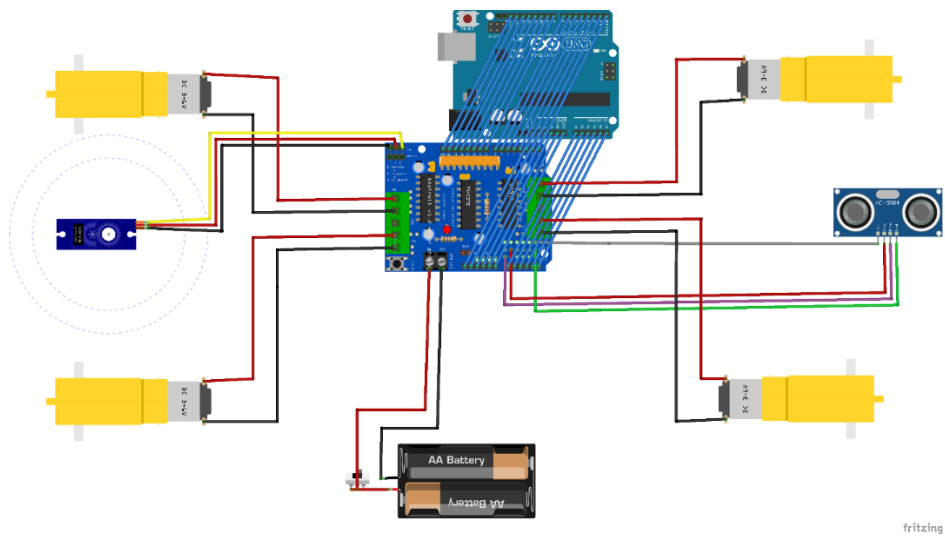


Figure 5: Circuit diagram

2.2.4. Schematic diagram

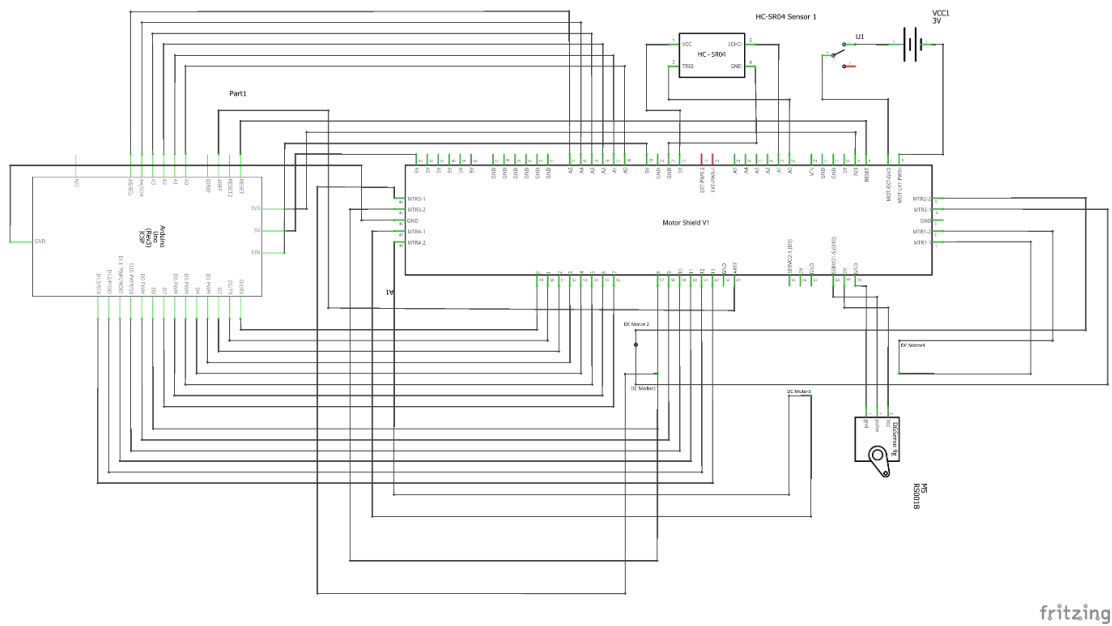


Figure 6: Schematic diagram

2.2.5. Block diagram

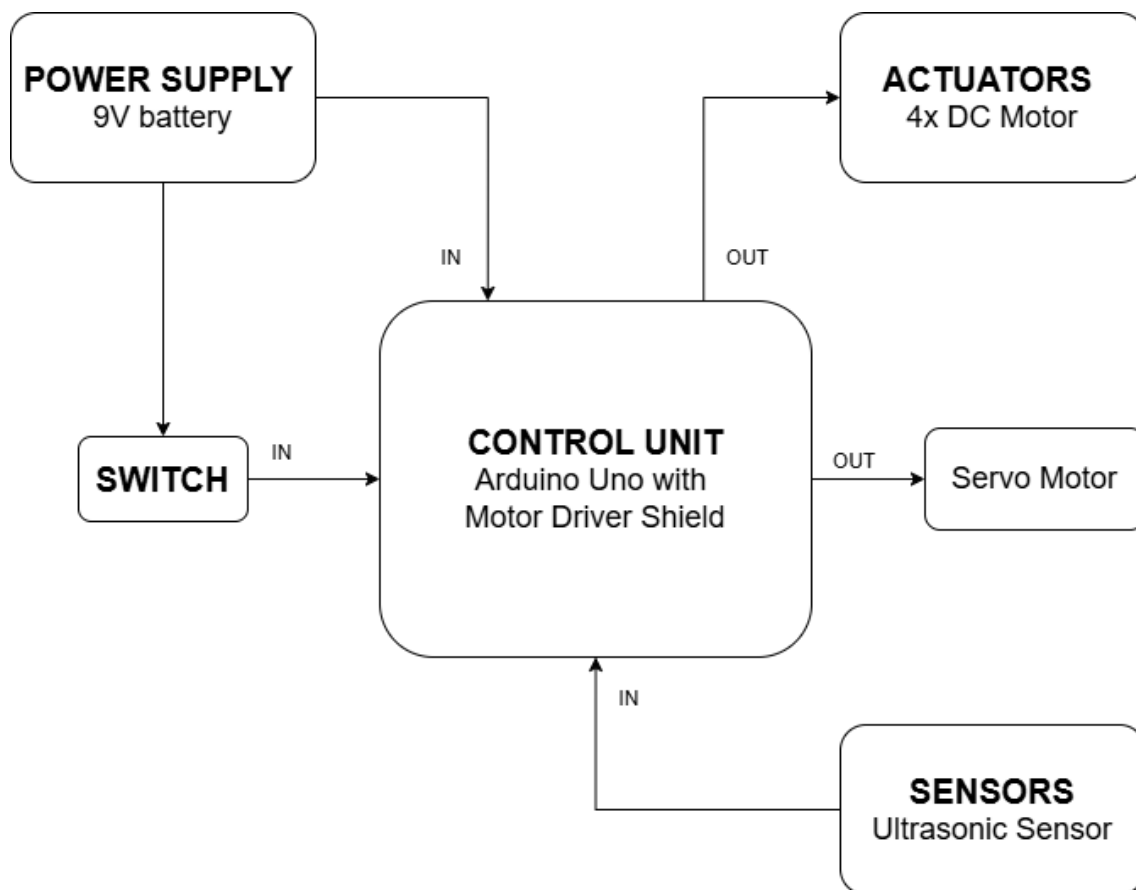


Figure 7: Block diagram

2.3. Requirement Analysis

2.3.1. Hardware

➤ Ultrasonic Sensors:

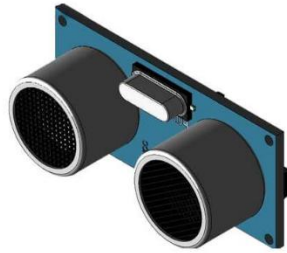


Figure 8: Ultrasonic Sensors

- **Description:** Outputs of energy in ultrasonic frequencies which enable object detection and measure the distance of the objects
- **Purpose:** Provides functionality for detecting obstacles and autonomous navigation. The sensors provide obstacle avoidance functionality which allows the vehicle to navigate through its environment (Jost, 2019).

➤ Wheel:



Figure 9: Wheel

- **Description:** Commonly a rotating circular component for controlling acceleration of a mobile robot.
- **Purpose:** The components serve as the basic appendages for locomotion to enable mobility on the robotic vehicle. They convert the energy produced by the motors into motion.

➤ **TT Motor:**



Figure 10: TT Motor

- **Description:** A small dynamo electric smooth-running motor (usually under load operates at 200 RPM at 3-6V) suitable for small robot driving mechanisms. Changes electrical power to rotational motion
- **Purpose:** Rotates the vehicle wheels. This is important for making commands given through the electronics responsive by physically advancing the vehicle in the directions specified and rotating it around its axis (Tricky Circuit, 2025).

➤ **Jumper Wires:**



Figure 11: Jumper Wires

- **Description:** Insulated Cables with solid tips, which allow simple plugging into breadboards and microcontroller pins, are available in male to male, male to female, and female to female.
- **Purpose:** Make electrical connection with various components such as sensors, motors and the Arduino. Jumper Wires make the system to be modular and offer easier troubleshooting or upgrading options (Hemmings, 2018).

➤ **Switch:**



Figure 12: Switch

- **Description:** Simple SPDT (Single Pole Double Throw) switch used for circuits to be switched ON/OFF or to toggle between two operating modes.
- **Purpose:** Start stop and reset the system manually. It is a simple control mechanism but adds an extra layer of safety and control to the system.

➤ **Arduino Uno:**



Figure 13: Arduino Uno

- **Description:** A microcontroller board with an ATmega328P at its core, featuring USB and serial communication alongside analog and digital I/O ports. It is open sourced.
- **Purpose:** The central processing unit. It will receive data from the sensors, process the programmed commands, and issue appropriate action signals for motors or displays as a output (Arduino, 2025).

➤ **Motor Driver Shield L293D:**

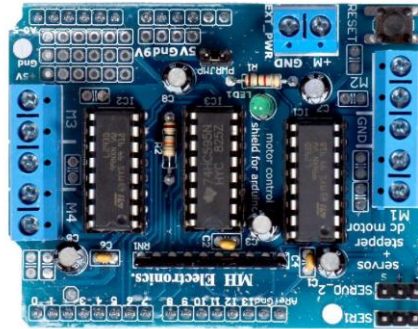


Figure 14: Motor Driver Shield L293D

- **Description:** Controls speed and direction of motors using signals from Arduino.
- **Purpose:** Enable supplying powers and control signals to the motors, while protecting the Arduino unit from excess current and permitting needed robots current functions safely (Electrical4u, 2024).

➤ **Servo Motor:**



Figure 15: Servo Motor

- **Description:** Provides precise control over angular position.
- **Purpose:** Supplies the required signals and the power to the DC motors and controls the speed of the motors and the signals provided. It also acts as a barrier between the Arduino and the high current (Electrical4U, 2024).

2.3.2. Software

➤ **Arduino IDE:**

Description: Arduino Integrated Development Environment (IDE) is an application to be run on a computer to develop, assemble, and load code into Arduino boards. It is open-source and freely downloadable software installable on Windows, Macintosh OSX, or Linux operating systems (Alkundi, 2025).



Figure 16: Arduino IDE

➤ **Fritzing:**

Description: Fritzing is an open-source platform that offer a software tool, a community website and services that are in line with Processing and Arduino that creates a creative environment which enables users to layout and produce expert PCBs, share and document their prototypes.



Figure 17 Fritzing

3. Development

3.1. Planning and design

The IoT-Enhanced Autonomous Arduino Car project's effective execution is based on the planning and design stage. The objective of the project must be specified precisely from the starting to direct any decisions that follow. The main objective of this project is to create an autonomous vehicle that can recognize obstacles and cleverly avoid them without assistance from a person, for this determining the intended working environments is also essential as it affects component selection and system resilience.

A vital phase in the design process is choosing the right components. The Arduino Uno Microcontroller was selected because of its strong community support, cost and adaptability. Ultrasonic sensors are employed for obstacle detection. A suitable motor driver is incorporated to effectively manage motor control and TT motors are used for locomotion to give sufficient speed and torque. Aside from being lightweight, the chassis is made to be strong enough to hold up all of hardware parts and remain stable when moving.

Finally, to make intelligent navigational decisions, such as halting or turning when obstacles are identified, the software design focuses on creating an obstacle avoidance system that evaluates real-time sensor data. To ensure safe navigation, this system dynamically regulates motor speed and direction. To verify the hardware and software integration, a preliminary prototype is built. The testing and improvement are then made to enhance the system's accuracy, responsiveness and dependability in various situations. This methodical planning and design procedure guarantees that the finished product successfully satisfies its functional and practical criteria.

3.2. Resource Collection

The resources needed to develop the IoT-enhanced autonomous Arduino car were collected from multiple sources to assemble all necessary component effectively. An

application letter was submitted to the college's IT resource department to request the available hardware. From the IT department, we received key components such as the Arduino Uno microcontroller, a motor driver, TT motor, jumper wires and wheels which are fundamental for building and controlling the car whereas the remaining materials like servo motor, switch and battery were sourced from local electronics stores. These components were carefully chosen to meet the functional and budgetary requirements of the project.

Finally, we gathered all essential parts required to develop the autonomous car system from the college's IT department and local suppliers, which ensured that both quality and availability were optimized, enabling smooth progress in assembling the hardware and programming the system for obstacle detection and avoidance.

3.3. System development

The system we have developed is an obstacle-avoiding car using Arduino UNO, which automatically stops, scans the area, and moves to an open place when an obstacle is detected.

Phase 1: Uploading the Code

We started by uploading the obstacle-avoidance code to the Arduino UNO via USB using a PC. The code was programmed to make the car stop when it detects an obstacle, scan the area using a servo-mounted sensor, and then move in a free direction. During this phase, we tested if the code was compiling and working properly.

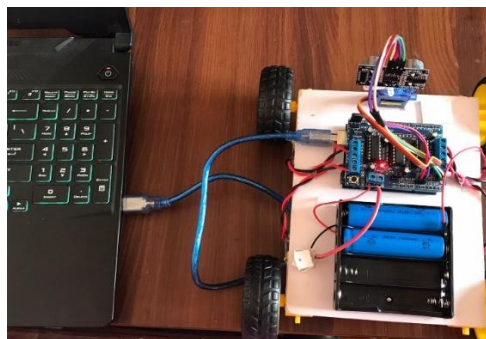


Figure 18: Code supplied to the Arduino

Phase 2: Ultrasonic Sensor Connection

Once the logic was in place, we integrated the ultrasonic sensor with the motor driver shield. The ultrasonic sensor is responsible for calculating distance based on echo time. The connections were made as follows:

VCC(Positive) to +5V on the motor driver shield

GND to GND on the motor driver shield

TRIG to Analog Pin A0

ECHO to Analog Pin A1

After establishing the connections, we ran multiple test scripts to confirm that accurate distance measurements were being received in real-time, which are critical for responsive obstacle detection.

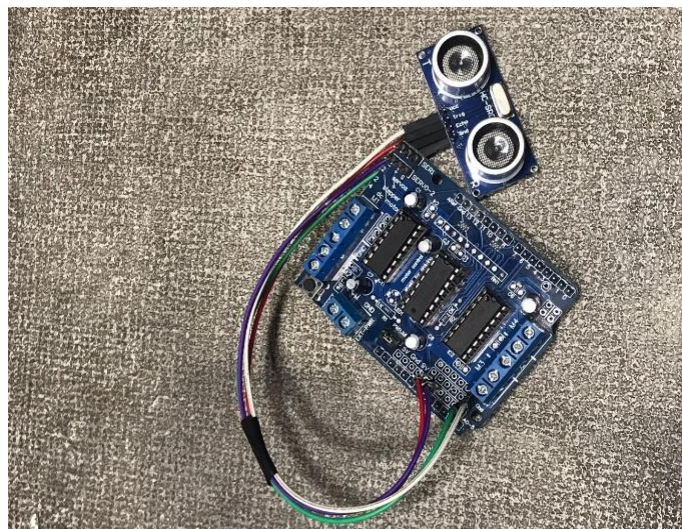


Figure 19: Ultrasonic sensor connected to the motor driver

Phase 3: Servo Motor Setup

After connecting the ultrasonic sensor to the motor driver shield, we connected the servo motor to the servo port (SERVO1) of the motor driver shield. This servo was mounted with the ultrasonic sensor to enable left and right scanning. The servo allows the sensor

to rotate and analyse the surroundings, enabling it to make smarter decisions by choosing the direction with the least obstruction.

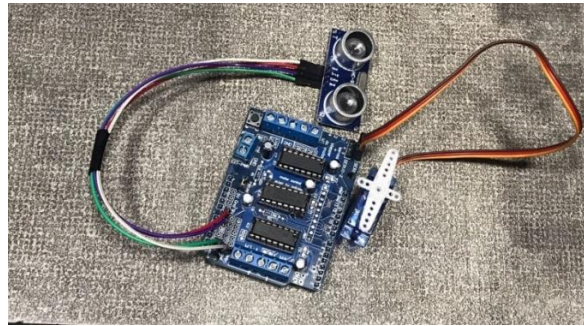


Figure 20: Servo motor connected to the servo port

Phase 4: Motor Connections

After connecting the servo motor, we connected the motors to the motor driver shield by the following connections:

Motor1 – M1

Motor2 – M2

Motor3 – M3

Motor4 – M4

Once the connections were made, we conducted a mobility test to check if all motors rotated in the correct direction and responded appropriately to control signals. We mounted all components securely onto a sun board chassis and added wheels to ensure stability and balance during movement.

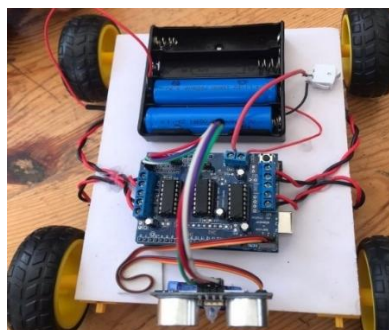


Figure 21: Motor connected to the motor driver shield

Phase 5: Power and Switch Integration

After assembling all the electronic and mechanical components, we then added two lithium batteries to provide sufficient power to the motors and the sensor. We then added a switch in the body of the car to control power delivery, allowing the system to be turned on/off without disconnecting wires.

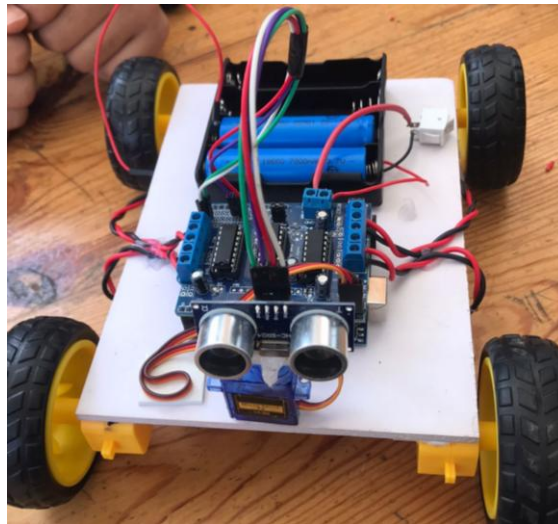


Figure 22: The battery and the switch were addressed to the bod

Phase 6: Real-World Testing

Finally, we tested the car in different environments. We observed its ability to detect obstacles and adjust its path. Minor code tweaks were made to improve accuracy and response time, ensuring the system works smoothly in real-world scenarios.

4. Results and Findings

The IoT-enhanced autonomous Arduino Car's development produced a several significant results and findings that demonstrates the effectiveness of the design and execution methods worked. Once the parts were put together and the control algorithms were programmed, the vehicle was put through a series of tests to evaluate its performance .

Testing

TEST 1: Testing to verify the execution of code.

Test	1
Objective	To compile the code and upload it to the Arduino after its compiled successfully.
Activity	- Arduino IDE compiles the code. - Arduino Uno is linked to a laptop to upload the code.
Expected Result	The code will compile and uploaded to Arduino without any issues.
Actual Result	The code was successfully uploaded and compiled.
Conclusion	The test was successful.

Table 1: Test 1

```

8
9
10 #include <AFMotor.h>
11 #include <NewPing.h>
12 #include <Servo.h>
13
14 #define TRIG_PIN A0
15 #define ECHO_PIN A1
16 #define MAX_DISTANCE 200
17 #define MAX_SPEED 100 // sets speed of DC motors
18 #define MAX_SPEED_OFFSET 20
19
20 NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE);
21
22 AF_DCMotor motor1(1, MOTOR12_1KHZ);
23 AF_DCMotor motor2(2, MOTOR12_1KHZ);
24 AF_DCMotor motor3(3, MOTOR34_1KHZ);
25 AF_DCMotor motor4(4, MOTOR34_1KHZ);
26 Servo myservo;
27
28 boolean goesforward=false;
29 int distance = 100;
30 int speedSet = 0;
31
32 void setup() {
33
34   myservo.attach(10);
35   myservo.write(115);
36   delay(2000);
37
38 }
39
40 void loop() {
41   // Read distance
42   distance = sonar.ping_cm();
43   // If distance is less than 20cm, stop the car and turn the servo
44   if (distance < 20) {
45     // Stop the car
46     motor1.setSpeed(0);
47     motor2.setSpeed(0);
48     motor3.setSpeed(0);
49     motor4.setSpeed(0);
50     // Turn the servo
51     myservo.write(115);
52     delay(2000);
53     // Change direction
54     goesforward = !goesforward;
55     // Set speed
56     speedSet = MAX_SPEED;
57   }
58   // Move the car
59   if (goesforward) {
60     motor1.setSpeed(speedSet);
61     motor2.setSpeed(speedSet);
62     motor3.setSpeed(-speedSet);
63     motor4.setSpeed(-speedSet);
64   } else {
65     motor1.setSpeed(-speedSet);
66     motor2.setSpeed(-speedSet);
67     motor3.setSpeed(speedSet);
68     motor4.setSpeed(speedSet);
69   }
70   delay(100);
71 }

```

avrdude: Device signature = 0x1e050f (probably m328p)
 avrdude: reading input file "C:\Users\ACER\AppData\Local\Arduino\sketches\BA351A099F007FD0D1C70F894AC3041A\gdi.ino.hex"
 avrdude: writing flash (3962 bytes):
 writing | ##### | 100% 0.56s
 avrdude: 3962 bytes of flash written
 avrdude done. Thank you.

Done uploading.

Figure 23: Successful execution of code

TEST 2: Testing to assure that obstacles are detected by the ultrasonic sensor.

Test	2
Objective	To verify that the ultrasonic sensor detects obstacles within its range.
Activity	- Place the autonomous car in an environment with no obstacles and gradually introduce objects Infront of the ultrasonic sensor. - The car will move backward and change the direction.
Expected Result	The ultrasonic sensor should detect any obstacles within its specified range.
Actual Result	The car read the distance and changed the direction.
Conclusion	The test was successful.

Table 2: Test 2

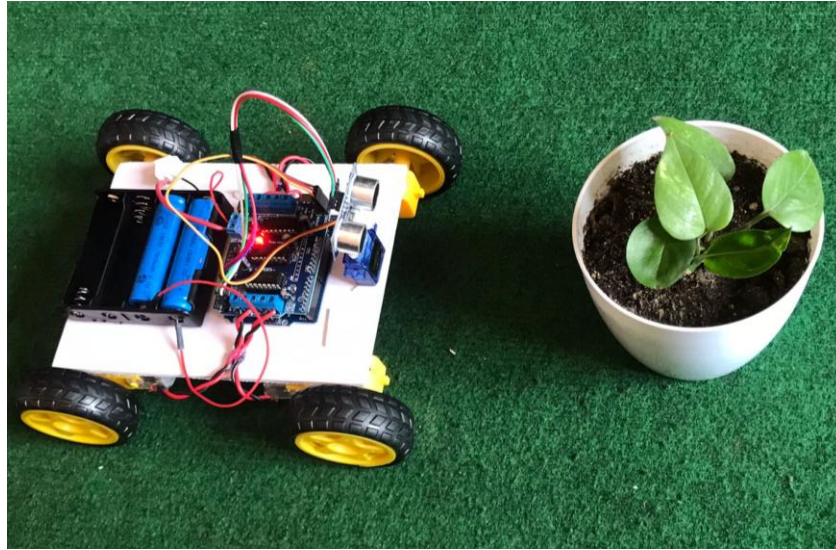


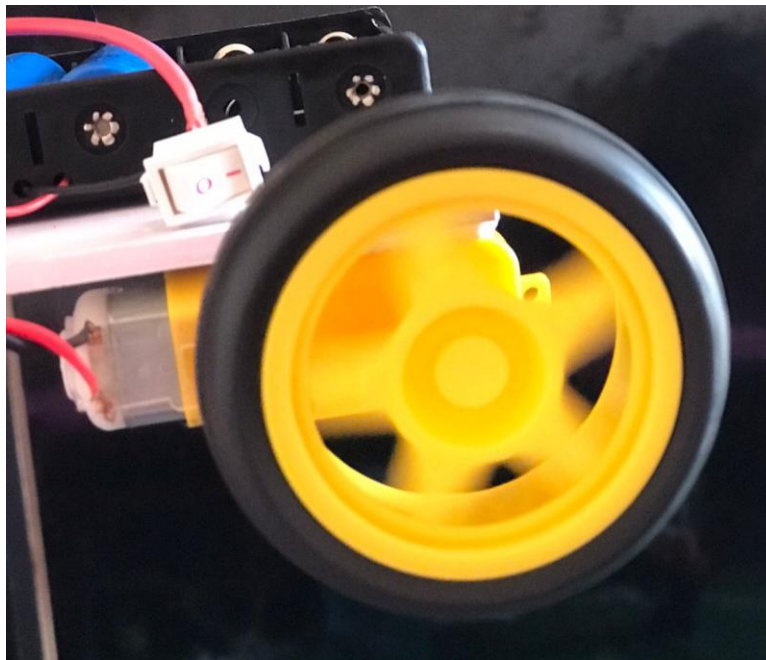
Figure 24: Obstacle in front



Figure 25: Direction changed

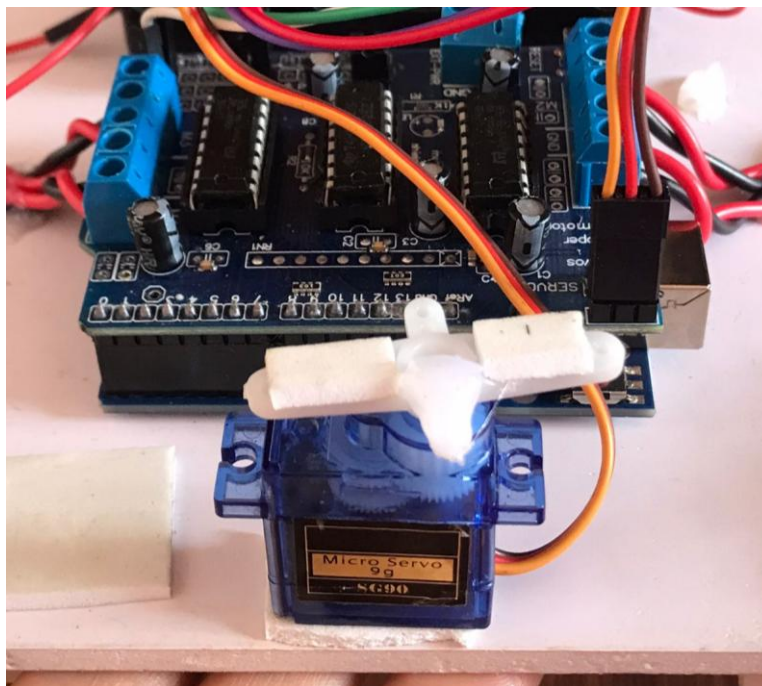
TEST 3: Testing to confirm that the motor rotates.

Test	3
Objective	To verify if the motor rotates.
Activity	- Connect the motor to the power supply and observe the motor's behaviour to confirm it starts rotating.
Expected Result	The wheel should start rotating after receiving the command.
Actual Result	The wheel started rotating and changed the direction of car.
Conclusion	The test was successful.

Table 3: Test 3*Figure 26: Rotating motor*

TEST 4: Testing the working mechanism of Servo Motor.

Test	4
Objective	To check the working of servo motor.
Activity	- Connect the servomotor to the microcontroller - Observe the servo motor's movement.
Expected Result	The servo motor should accurately move and hold its position firmly once the target angle is reached.
Actual Result	The servo motor was moved smoothly.
Conclusion	The test was successful.

Table 4: Test 4*Figure 27: Before connecting to the motor driver shield*

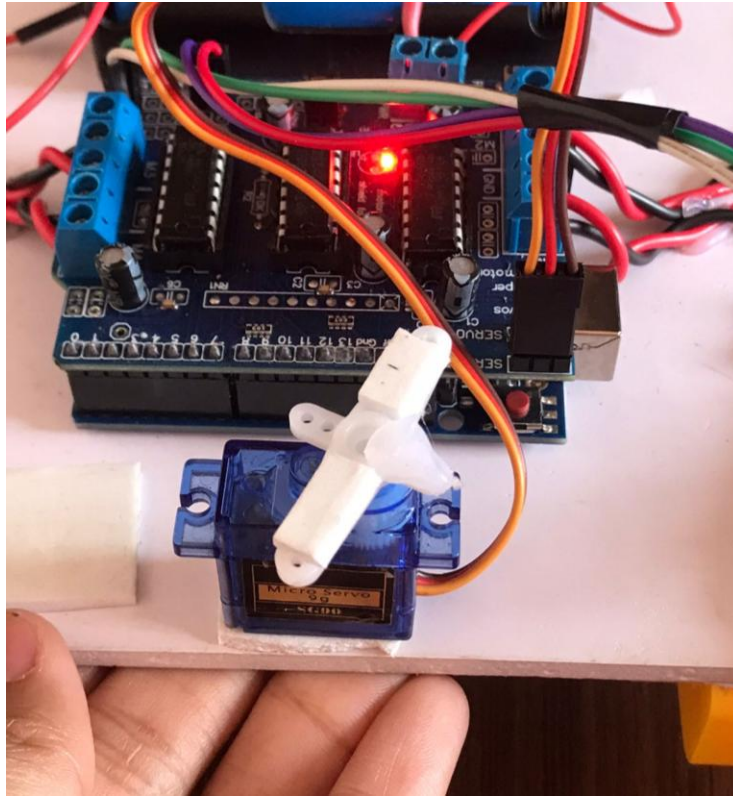
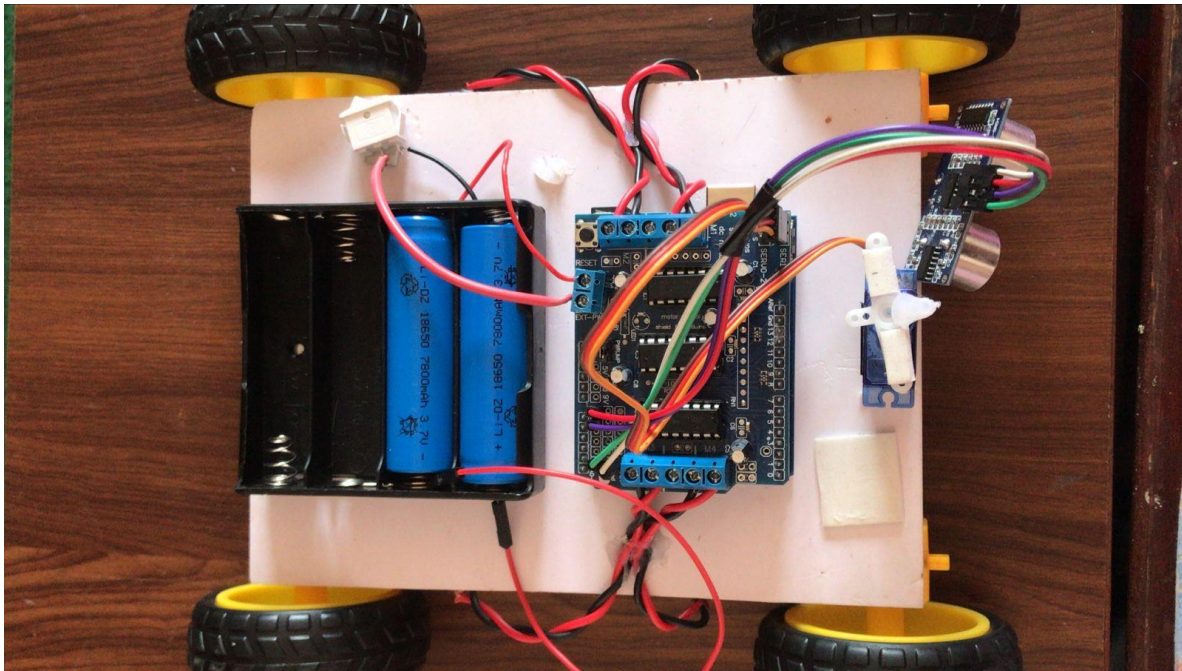


Figure 28: Direction changed after the connection

TEST 5: Testing the working mechanism of motor driver shield.

Test	5
Objective	To check the working mechanism of motor driver shield.
Activity	- After the code is uploaded successfully , the motor driver shield should supply the power.
Expected Result	The motor driver shield should supply the power.

Actual Result	The motor driver shield supplied the power.
Conclusion	The test was successful.

Table 5: Test 5*Figure 29: Before power supplied*

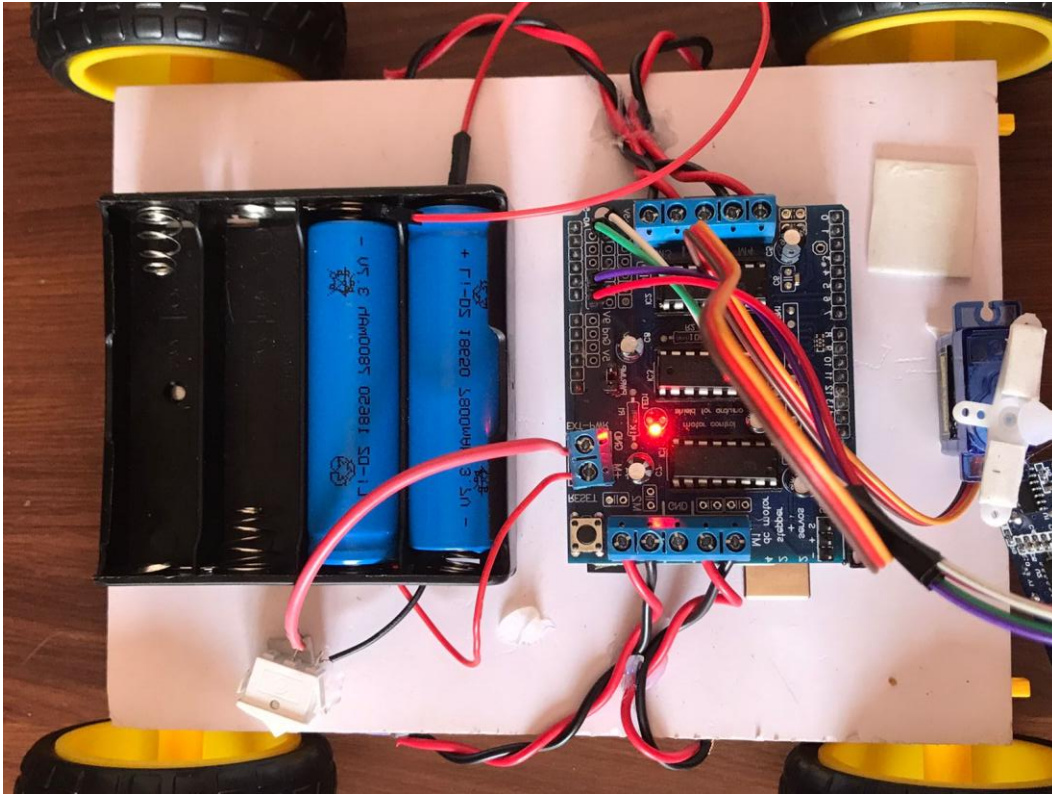


Figure 30: After power is supplied

Test 6: Testing power supply with Lithium Batteries

Test	6
Objective	To test the system's response to different power levels using lithium batteries.
Activity	- Connected 4 lithium batteries to power the system; wires began to heat, melt, and produce smoke. - Tried again with 3 batteries and observed the same issue.

	- Finally, powered the system using only 2 batteries, and it worked correctly.
Expected Result	The system should function correctly with stable power and no overheating.
Actual Result	Using 4 or 3 batteries caused overheating and wire damage; 2 batteries provided stable operation.
Conclusion	The system draws more current than required with 3 or 4 batteries, leading to overheating. Two batteries are optimal for safe operation.

Table 6: Test 6



Figure 31: Used 3 batteries



Figure 32: Wire damaged

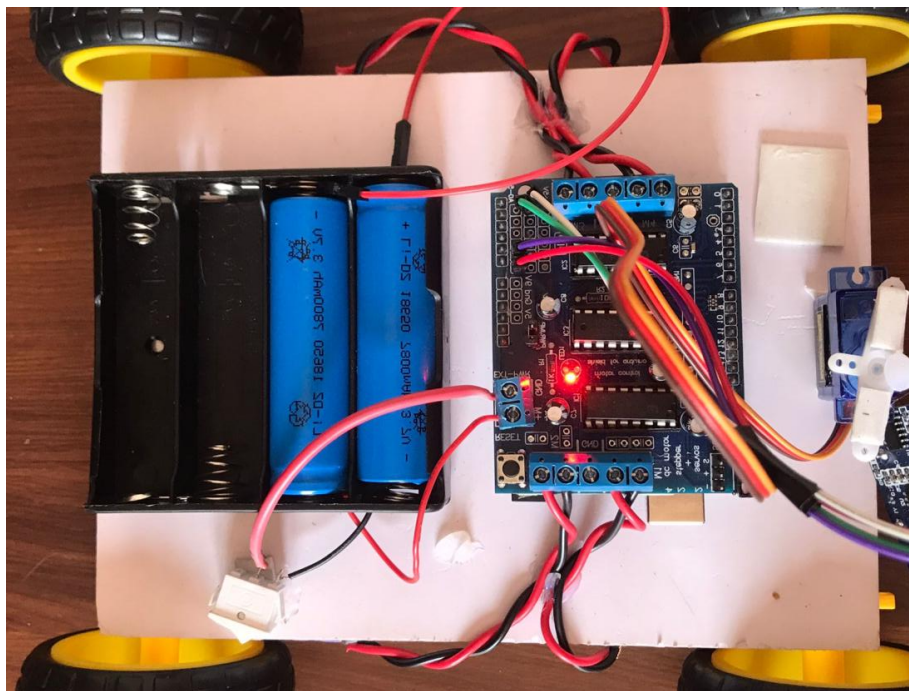


Figure 33: After using 2 batteries

5. Future works

This vehicle is still in its beta form and has a lot of room for improvement. Integrating Artificial Intelligence into the vehicle can increase its functionality, as AI has the potential of detecting people, objects, and vehicles more accurately with the traffic surveillance systems while choosing the safest route to avoid obstacles. With the integration of an AI powered camera, the ability to 'see' would increase, thus making driving more efficient and 'safer'.

Users could control the vehicle through the mobile application that we plan to design, making vehicle usage more enjoyable and accessible. Features would include starting or stopping the vehicle, checking the battery's status, route summoning, or setting up alerts for malfunctions. Such features would allow for better interaction with the vehicle.

Additions could be made so the vehicle could autopilot basics functions such as self-parking, tracking, and environment scanning for changes like traffic or weather in real time. This would be practical for aiding services to the elderly or disabled people who are prone to requiring extra assistance.

To make the vehicle run more smoothly and use less energy, we will put new tech in the vehicle's motor so it could run more smoothly and efficiently, therefore becoming more eco-friendly.

6. Conclusion

The IoT project using ultrasonic sensors dubbed as “Autonomous Arduino Car with Object Detection and Avoidance” has been a fruitful endeavour and has given a deeper insight into IoT and embedded systems. Using the ultrasonic sensors along with an Arduino microcontroller, we as a group were able to create a functioning car that could detect obstacles in its path and change directions to avoid collisions completely on its own.

Throughout the development process, we focused on the operation of the hardware and software working seamlessly regarding one another. Realtime data from the sensors allow the system to make quick on the fly decisions. With rigorous testing and troubleshooting, we were able to perk up its performance significantly from the first prototype. We encountered certain challenges like the structure of the car being unlevelled, sensor precision and motor control. These challenges helped in understanding the need for refinement of both the physical setup and the code in IoT projects. In the future, there is still a lot of room for improvement. Updated versions could have more advanced sensors i.e.. Lidar or infrared for better accuracy. These enhancements could lead to real world applications in different sectors such as warehouse automation, smart vehicles, so on and so forth.

Overall, this project gave us a glimpse into the vast world of IoT and robotics. It is a great launch point for anyone to get started in IoT. The combined research, programming and handwork efforts of the entire group was highly important in bringing this project to life.

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8. Appendix

8.1. Source code

```
#include <AFMotor.h>
#include <NewPing.h>
#include <Servo.h>

#define TRIG_PIN A0
#define ECHO_PIN A1
#define MAX_DISTANCE 200
#define MAX_SPEED 100 // sets speed of DC motors
#define MAX_SPEED_OFFSET 20

NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE);

AF_DCMotor motor1(1, MOTOR12_1KHZ);
AF_DCMotor motor2(2, MOTOR12_1KHZ);
AF_DCMotor motor3(3, MOTOR34_1KHZ);
AF_DCMotor motor4(4, MOTOR34_1KHZ);
Servo myservo;

boolean goesForward=false;
int distance = 100;
int speedSet = 0;

void setup() {

  myservo.attach(10);
```

```
myservo.write(115);  
delay(2000);  
distance = readPing();  
delay(100);  
distance = readPing();  
delay(100);  
distance = readPing();  
delay(100);  
distance = readPing();  
delay(100);  
}
```

```
void loop() {  
  int distanceR = 0;  
  int distanceL = 0;  
  delay(40);  
  
  if(distance<=20)  
  {  
    moveStop();  
    delay(100);  
    moveBackward();  
    delay(300);  
    moveStop();  
    delay(200);  
    distanceR = lookRight();  
    delay(200);  
    distanceL = lookLeft();  
  }
```

```
    delay(200);

    if(distanceR>=distanceL)
    {
        turnRight();
        moveStop();
    }else
    {
        turnLeft();
        moveStop();
    }
    }else
    {
        moveForward();
    }
    distance = readPing();
}

int lookRight()
{
    myservo.write(50);
    delay(500);
    int distance = readPing();
    delay(100);
    myservo.write(115);
    return distance;
}
```

```
int lookLeft()
{
    myservo.write(170);
    delay(500);
    int distance = readPing();
    delay(100);
    myservo.write(115);
    return distance;
    delay(100);
}
```

```
int readPing() {
    delay(70);
    int cm = sonar.ping_cm();
    if(cm==1)
    {
        cm = 250;
    }
    return cm;
}
```

```
void moveStop() {
    motor1.run(RELEASE);
    motor2.run(RELEASE);
    motor3.run(RELEASE);
    motor4.run(RELEASE);
}
```

```
void moveForward() {

    if(!goesForward)
    {
        goesForward=true;
        motor1.run(FORWARD);
        motor2.run(FORWARD);
        motor3.run(FORWARD);
        motor4.run(FORWARD);

        for (speedSet = 0; speedSet < MAX_SPEED; speedSet +=2) // slowly bring the speed
        up to avoid loading down the batteries too quickly
        {
            motor1.setSpeed(speedSet);
            motor2.setSpeed(speedSet);
            motor3.setSpeed(speedSet);
            motor4.setSpeed(speedSet);
            delay(5);
        }
    }
}

void moveBackward() {
    goesForward=false;
    motor1.run(BACKWARD);
    motor2.run(BACKWARD);
    motor3.run(BACKWARD);
    motor4.run(BACKWARD);

    for (speedSet = 0; speedSet < MAX_SPEED; speedSet +=2) // slowly bring the speed
    up to avoid loading down the batteries too quickly
```

```
{  
  motor1.setSpeed(speedSet);  
  motor2.setSpeed(speedSet);  
  motor3.setSpeed(speedSet);  
  motor4.setSpeed(speedSet);  
  delay(5);  
}  
}
```

```
void turnRight() {  
  motor1.run(FORWARD);  
  motor2.run(FORWARD);  
  motor3.run(BACKWARD);  
  motor4.run(BACKWARD);  
  {  
    motor1.setSpeed(120);  
    motor2.setSpeed(120);  
    motor3.setSpeed(120);  
    motor4.setSpeed(120);  
  }  
  delay(700);  
  motor1.run(FORWARD);  
  motor2.run(FORWARD);  
  motor3.run(FORWARD);  
  motor4.run(FORWARD);  
}
```

```
void turnLeft() {
```

```

motor1.run(BACKWARD);
motor2.run(BACKWARD);
motor3.run(FORWARD);
motor4.run(FORWARD);
{
  motor1.setSpeed(120);
  motor2.setSpeed(120);
  motor3.setSpeed(120);
  motor4.setSpeed(120);
}
delay(700);
motor1.run(FORWARD);
motor2.run(FORWARD);
motor3.run(FORWARD);
motor4.run(FORWARD);
}

```

8.2. Individual contribution plan

We have a team of 5 members that worked together to build Autonomous Arduino Car with Object Detection and Avoidance. The roles and contribution of every team member have been listed below:

Student Name	Role	Contribution
Manita Basnet	- Introduction - Development part - Formatting & review of the project - Device integration and arrangement.	20%

	- Debugging of code and overall integration of the project - Testing	
Krish Shrestha	- System architecture and Schematic diagram - Development part - Idea and research of the project - Program coding and integration - Debugging	20%
Mohammad Masood Siddiqui	- Background(Requirement analysis) - Hardware software evaluate. - Overall project integration - Future works - Block diagram	20%
Mishan Katuwal Chettri	- Research of the project - Abstract and acknowledgement. - Hardware evaluation. - Resource management.	20%
Kushal Maharjan	- Overall conclusion of the project - Design of the project - Review of the project - Flowchart - Resource management	20%

Table 7: Individual Contribution Plan